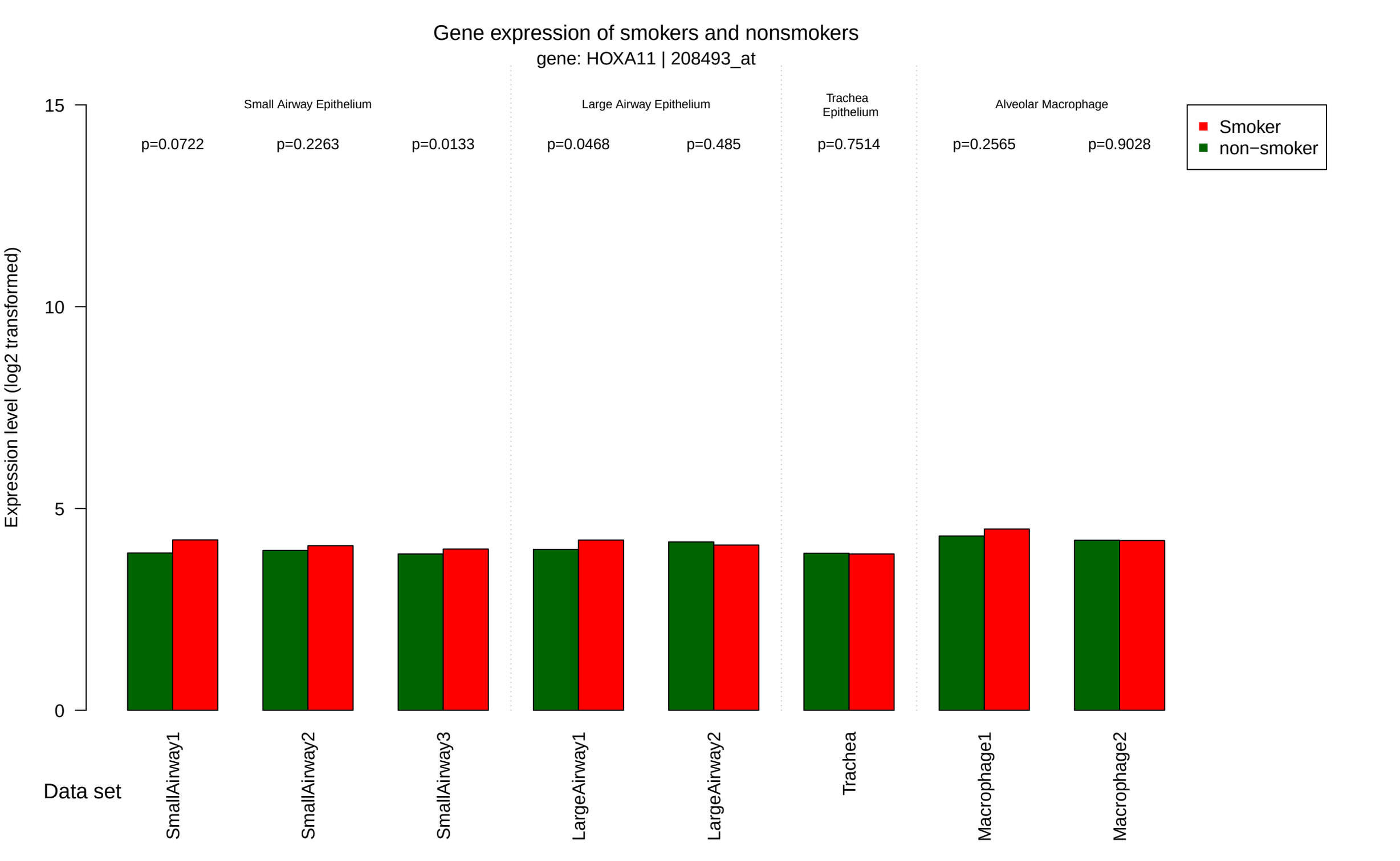
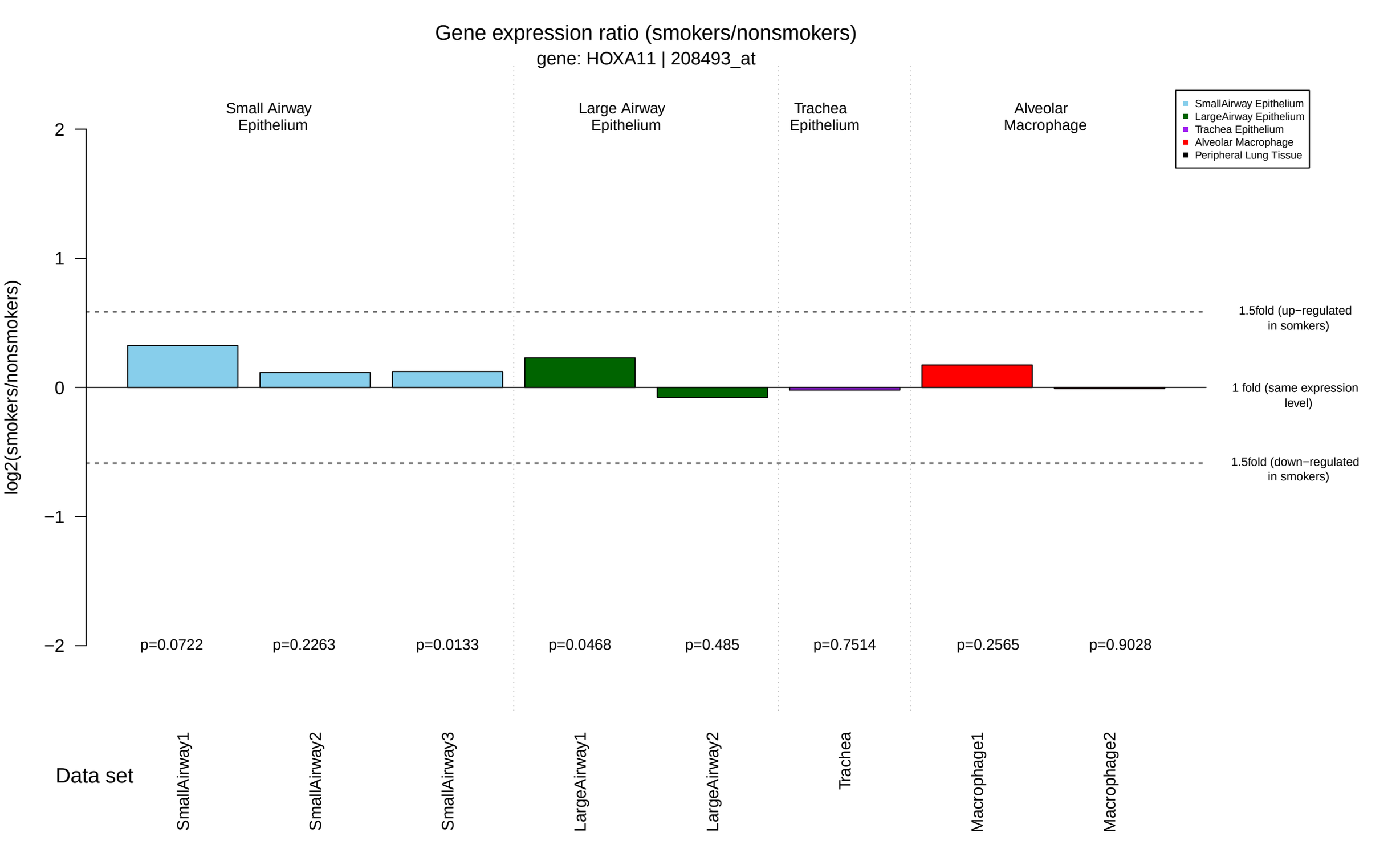
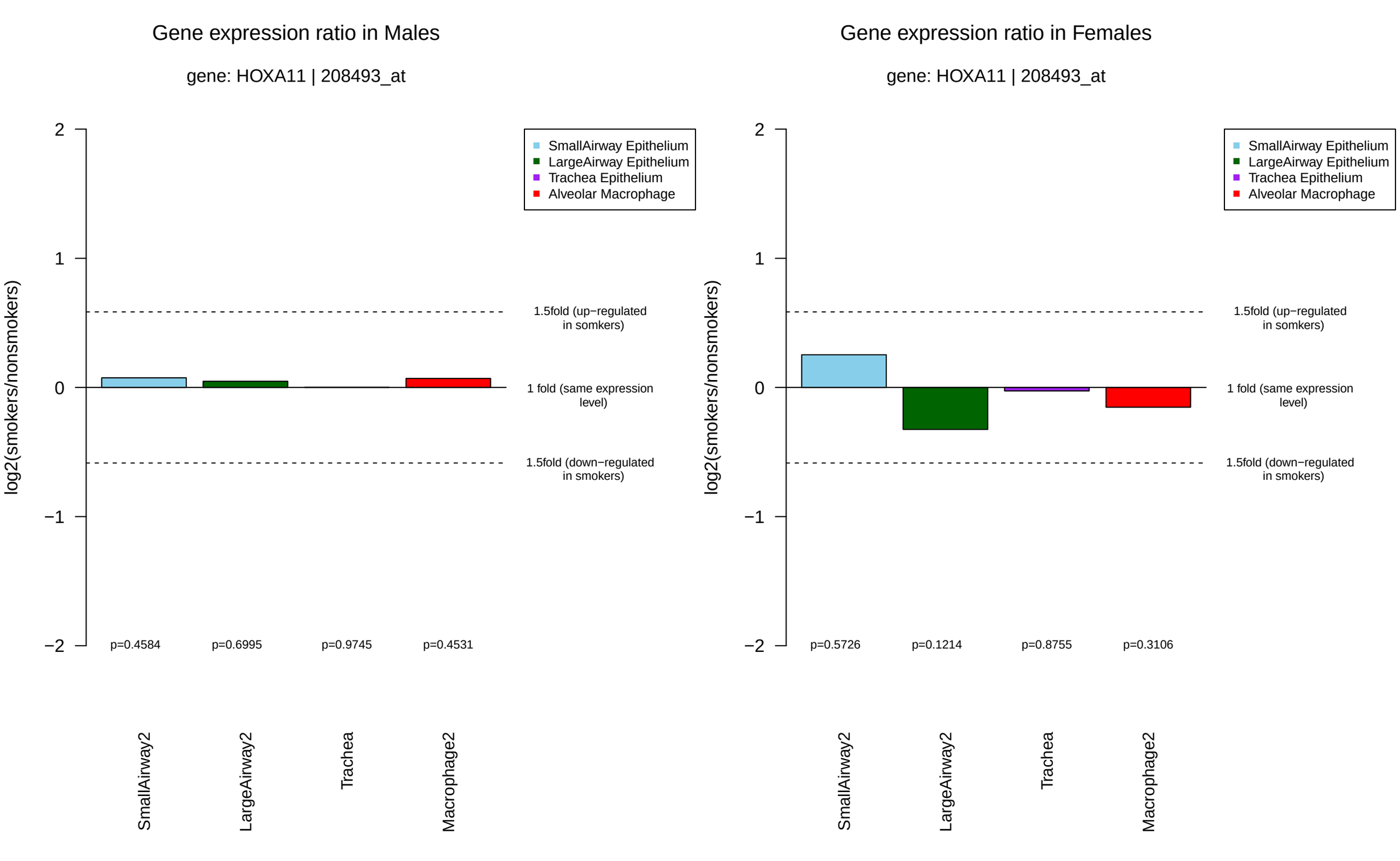
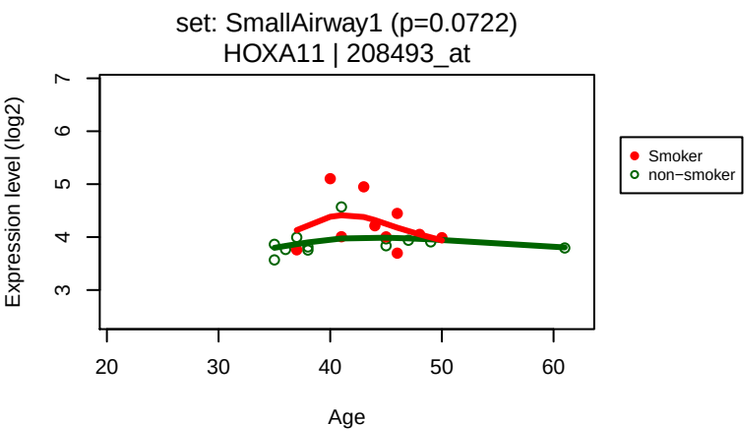
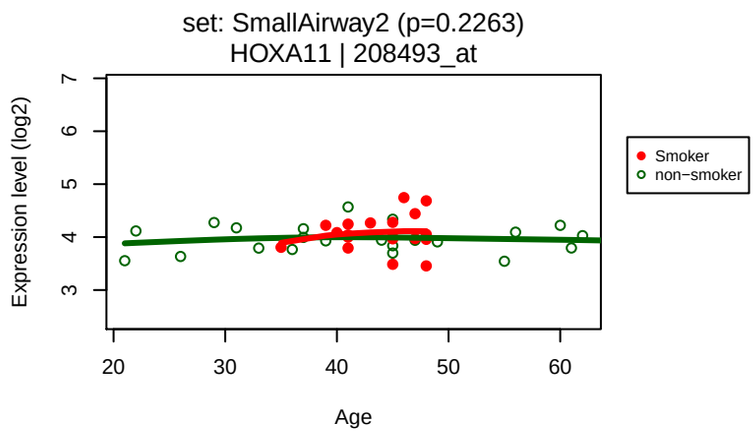
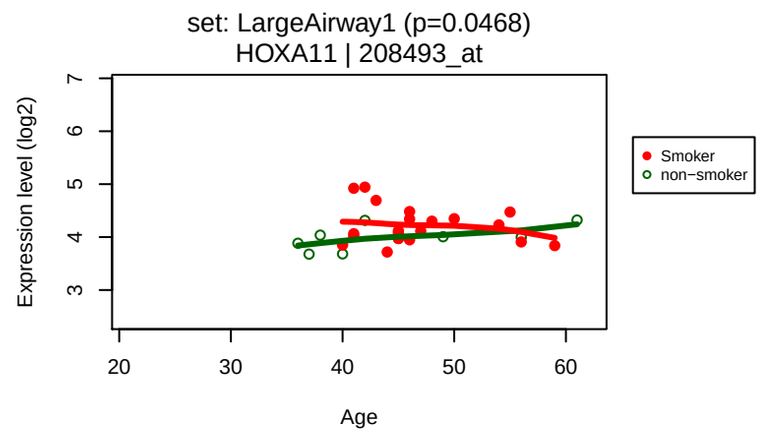

Gene expression of smokers and nonsmokers
gene: HOXA11 | 208493_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0722
p=0.2263
p=0.0133
p=0.0468
p=0.485
p=0.7514
p=0.2565
p=0.9028
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

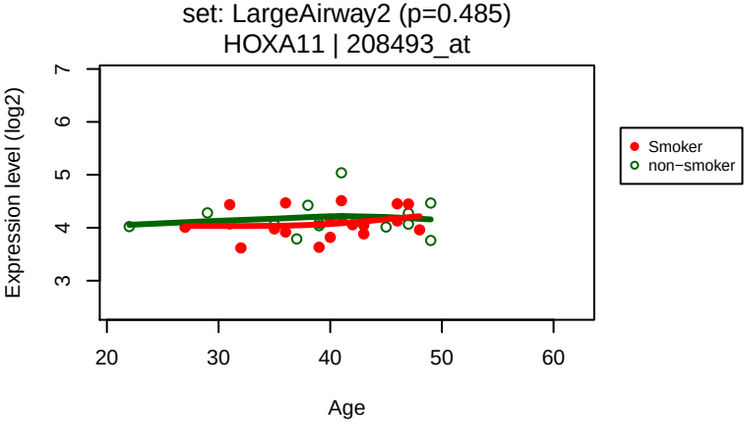

Gene expression ratio (smokers/nonsmokers)
gene: HOXA11 | 208493_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0722
p=0.2263
p=0.0133
p=0.0468
p=0.485
p=0.7514
p=0.2565
p=0.9028
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

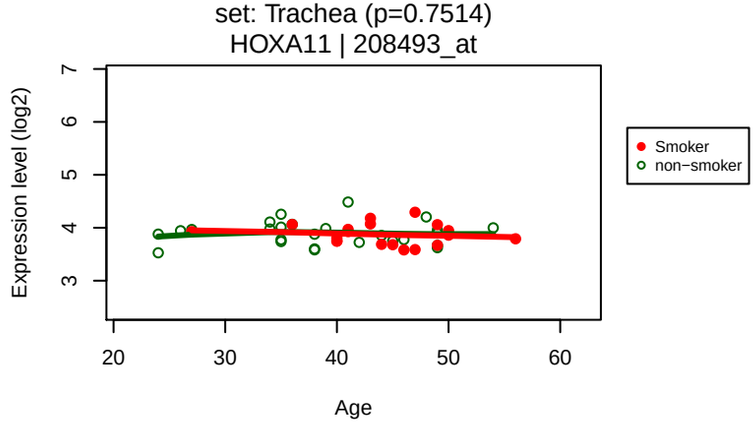

Gene expression ratio in Males
gene: HOXA11 | 208493_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4584
p=0.6995
p=0.9745
p=0.4531
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HOXA11 | 208493_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5726
p=0.1214
p=0.8755
p=0.3106
SmallAirway2
LargeAirway2
Trachea
Macrophage2

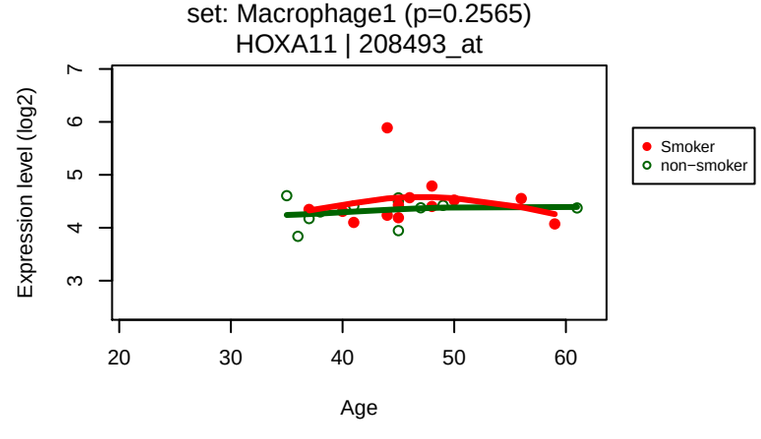

set: SmallAirway1 (p=0.0722)
HOXA11 | 208493_at
Expression level (log2)
Age
Smoker
non−smoker

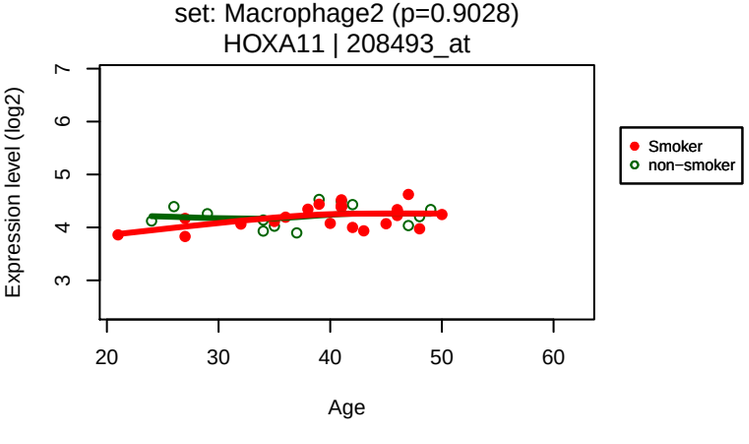

set: SmallAirway2 (p=0.2263)
HOXA11 | 208493_at
Expression level (log2)
Age
Smoker
non−smoker

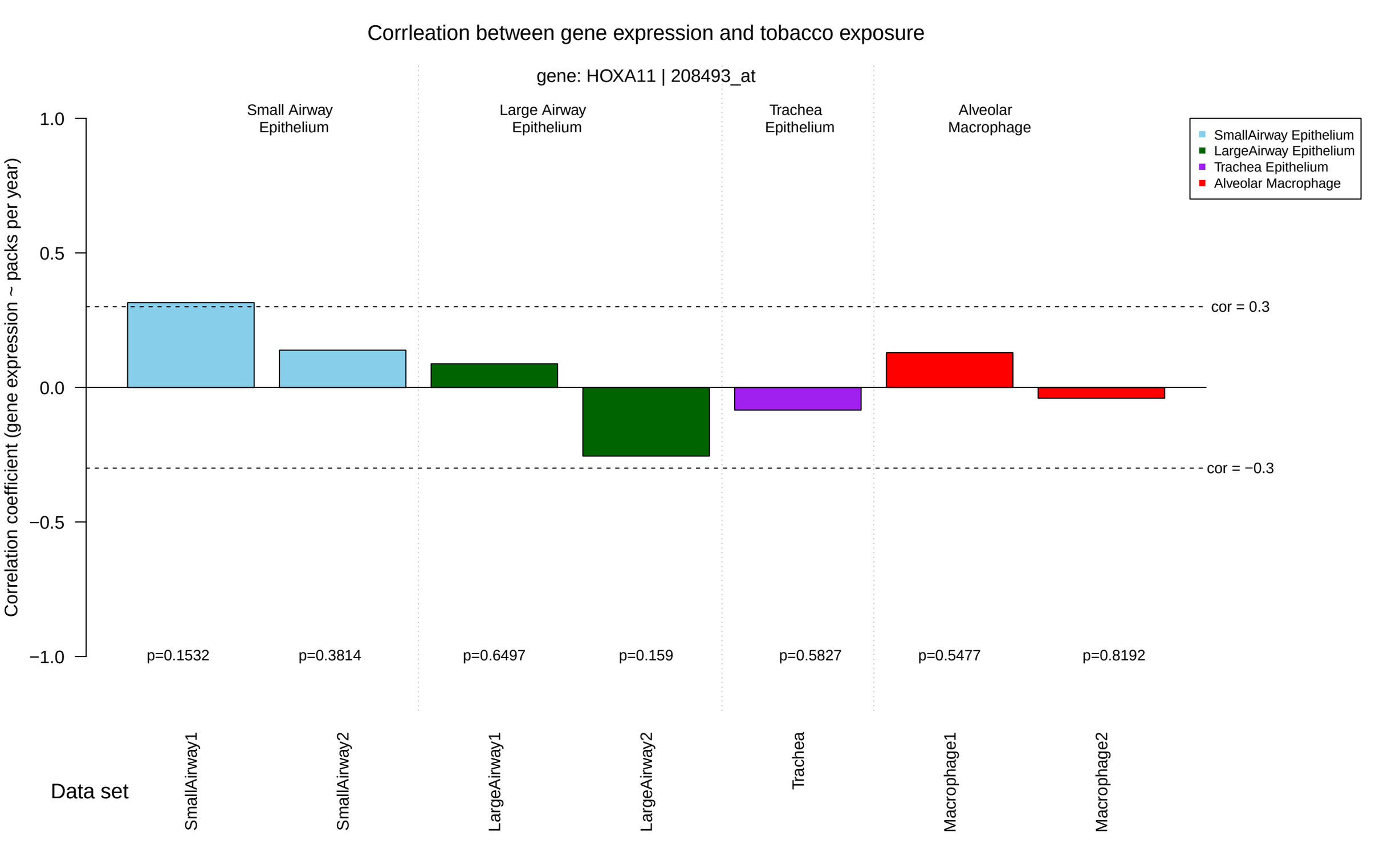

set: LargeAirway1 (p=0.0468)
HOXA11 | 208493_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.485)
HOXA11 | 208493_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7514)
HOXA11 | 208493_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2565)
HOXA11 | 208493_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9028)
HOXA11 | 208493_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HOXA11 | 208493_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1532
p=0.3814
p=0.6497
p=0.159
p=0.5827
p=0.5477
p=0.8192
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2